Relativity

Inertial Frame:

The frames with respect to which an unaccelerated body appears unaccelerate are inertial frames. In other words, the frames which are at rest or in uniform translatory motion relative to one another are inertial frames. In brief, we may say "An inertial frame is one in which Newton's first law is true"

Non-inertial Frame:

The frames with respect to which an unaccelerated body appears accelerated are called non-inertial frames. In other words, the accelerated frames are non-inertial.

Postulates of Special theory of Relativity:

The special theory of relativity is based on the two postulates.

Postulate-I:

The fundamental laws of physics have the same form for all inertial system i.e., for all reference systems at rest or moving with constant linear velocity relative to one another.

Postulate-II:

The velocity of light in free space has the same value for all inertial observers.

Lorentz Transformation Equations:

$$x' = \frac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

$$y' = y \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

$$z' = z \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (3)$$

$$t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (4)$$

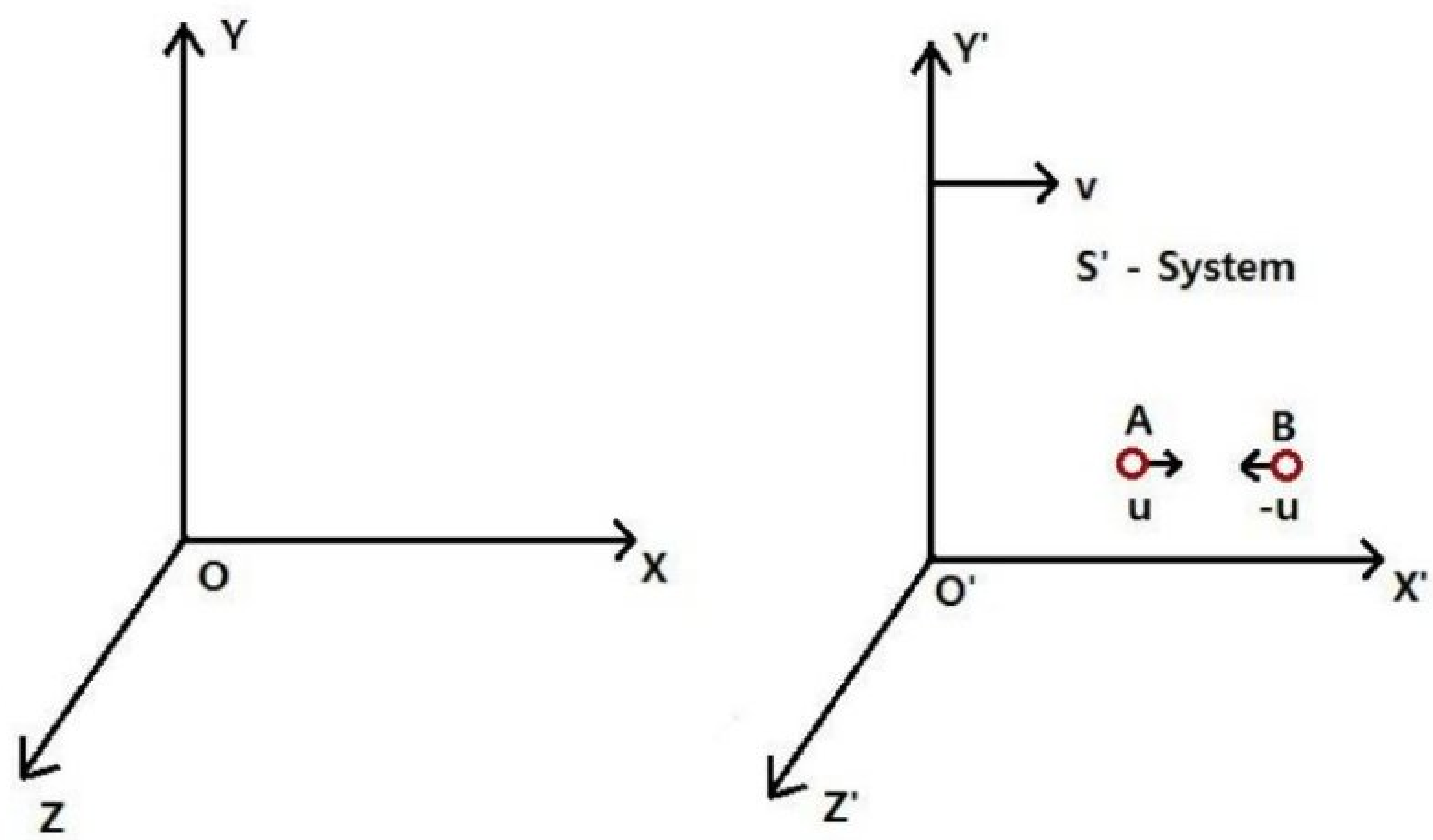
Variation of Mass with Velocity:

Derivation: Consider two systems *S* and *S'*. *S'* is moving with a constant velocity v relative to the system *S,* in the positive X-direction)Fig: 1(. Suppose that in the system *S'*, two exactly similar elastic balls *A* and *B* approach each other at equal speeds)i.e., *u* and —*u*(. Let the mass of each ball be *m* in *S'*. They collide with each other and after collision coalesce into one body. According to the law, of conservation of momentum,

Momentum of ball A + momentum of ball B = momentum of coalesced mass.

Or *mu* +)— *mu*(=- momentum of coalesced mass = 0.

Thus, the coalesced mass must be at rest in *S'* system.



Let us now consider the collision with reference to the system *S*.

Let u_1, and u_2 be the velocities of the balls relative to *S*. Then,

$u_1 = \frac{u+v}{l+\frac{uv}{c^2}}$ ………………………………………….(1)

and $u_2 = \frac{-u+v}{l-\frac{uv}{c^2}}$…………………………………….(2)

After collision, velocity of the coalesced mass is v relative to the system S.

Let mass of the ball A travelling with velocity u_1 be m_1 and that of B with velocity u_2 be m_2 in the system S. Total momentum of the balls is conserved. Therefore,

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) v \qquad ..(3)$$

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Substituting for u_1 and u_2 form (1) and (2), we have ,

$$m_1\left[\frac{u+v}{1+\frac{uv}{c^2}}\right]+m_2\left[\frac{-u+v}{1-\frac{uv}{c^2}}\right]=(m_1+m_2)\text{v}$$

Or $$m_1\left[\frac{u+v}{1+\frac{uv}{c^2}}-v\right]=m_2\left[-v-\frac{-u+v}{1-\frac{uv}{c^2}}\right]$$

Or $$m_1\left[\frac{u+v-v-\frac{uv^2}{c^2}}{1+\frac{uv}{c^2}}\right]==m_2\left[\frac{v-\frac{uv^2}{c^2}+u-v}{1-uv/c^2}\right]$$

Or $$m_1\left[\frac{u\left(1-\frac{v^2}{c^2}\right)}{1+\frac{uv}{c^2}}\right]=m_2\left[\frac{u\left(1-\frac{v^2}{c^2}\right)}{1-\frac{uv}{c^2}}\right]$$

Or $$\frac{m_1}{m_2}=\frac{1+\frac{uv}{c^2}}{1-\frac{uv}{c^2}} \qquad ..(4)$$

Also $$1\text{-}\frac{u^2}{c^2}=1-\frac{\left\{\frac{u+v}{c}\right\}^2}{\left\{1+\frac{uv}{c}\right\}^2}$$

$$=\frac{1+\frac{u^2v^2}{c^2}+\frac{2uv}{c^2}-\frac{u^2}{c^2}-\frac{v^2}{c^2}-\frac{2uv}{c^2}}{(1+\frac{uv}{c^2})^2}$$

$$=\frac{(1+\frac{u^2}{c^2})-\frac{v^2}{c^2}-(1-\frac{u^2}{c^2})}{(1+\frac{uv}{c^2})^2}$$

$$1-\frac{u1^2}{c^2}=\frac{\left(1-\frac{u^2}{c^2}\right)\left(1-\frac{v^2}{c^2}\right)}{(1+\frac{uv}{c^2})^2} \qquad ..(5)$$

Similarly, $$1-\frac{u2^2}{c^2}=\frac{\left(1-\frac{u^2}{c^2}\right)\left(1-\frac{v^2}{c^2}\right)}{(1+\frac{uv}{c^2})^2} \qquad ..(6)$$

Dividing equation (6) by equation (5),

$$\frac{1-\frac{{u_2}^2}{c^2}}{1-\frac{{u_1}^2}{c^2}}=\frac{(1+\frac{uv^2}{c^2})^2}{(1-\frac{uv^2}{c^2})^2}$$

Or $$\frac{\sqrt{1-\frac{{u_2}^2}{c^2}}}{\sqrt{1-\frac{{u_1}^2}{c^2}}}=\frac{1+\frac{uv^2}{c^2}}{1-\frac{uv^2}{c^2}}\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots.(7)$$

From equations (7) and (4),

$$\frac{m_1}{m_2} = \frac{\sqrt{1-\frac{u_2{}^2}{c^2}}}{\sqrt{1-\frac{u_1{}^2}{c^2}}}$$

Or $$m_1\sqrt{1-\frac{u_1{}^2}{c^2}}=m_2\sqrt{1-\frac{u_2{}^2}{c^2}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (8)$$

Since the L.H.S. and R.H.S. of equation (8) are independent of one another, the above result can be true only if each is a constant. Therefore.

$$m_1\sqrt{1-\frac{u_1{}^2}{c^2}}=m_2\sqrt{1-\frac{u_2{}^2}{c^2}} = m_0$$

The constant denoted by $m_0$is called the *rest mass* of the body and corresponds to *zero* velocity

Thus, $$m_1=\frac{m_0}{\sqrt{1-\frac{u_1{}^2}{c^2}}}$$

In general, if *m* denotes the mass of a body when it is moving with a velocity *v*. then,

$$\boldsymbol{m=\frac{m_0}{\sqrt{1-v^2/c^2}}}$$

This is the relativistic formula for the variation of mass with velocity.

If we put $v \to c$ in equation (9), we have $m \to \infty$ i.e., an object travelling at the speed of light would have infinite mass. Thus, Eqn. (9) shows that no material body can have a velocity equal to, or greater than the velocity of light.

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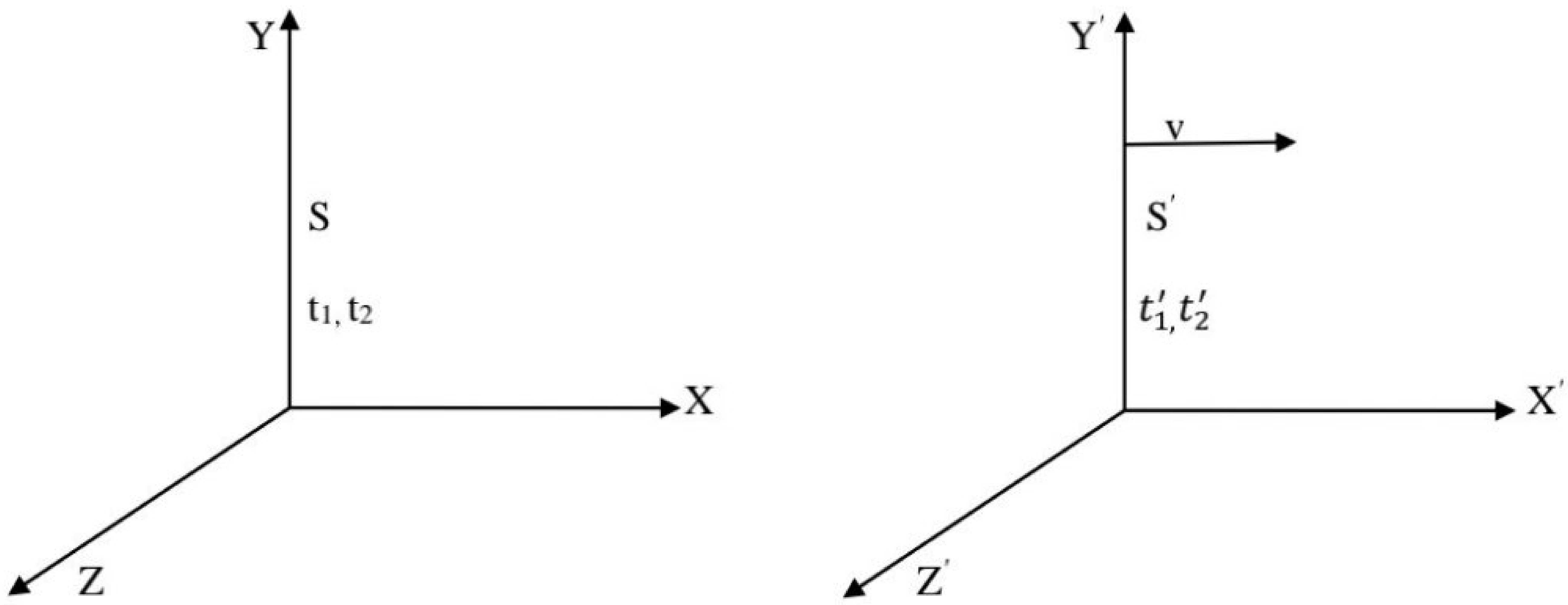
Time Dilation:

Let us consider two frames of reference S and S'. S' is moving with velocity v along to +ve direction of x-axis relative to S.



Let, a clock be placed in reference system S which is at rest gives a signal at time t_1 in system S and suppose that t_1' is the time measured by the observer in S', corresponding to the time t_1. According to Lorentz transformation equations, we have,

$$t_1' = \frac{t_1 - \frac{vx}{c^2}}{\sqrt{1-\frac{v^2}{c^2}}} \quad \text{……………………………(1)}$$

If this clock again gives a signal at time t_2, then corresponding time t_2' in frame S' is given by,

$$t_2' = \frac{t_2 - \frac{vx}{c^2}}{\sqrt{1-\frac{v^2}{c^2}}} \quad \text{……………………………(2)}$$

Therefore, the clock gives signals at an interval $(t_2 - t_1) = \Delta t$ in system S, when this interval is measured from the moving system it is equal to $t_2' - t_1' = \Delta t'$.

Thus, from equations 1 and 2, we have

$$t_2' - t_1' = \Delta t' = \frac{t_2 - t_1}{\sqrt{1-\frac{v^2}{c^2}}}$$

$$\therefore \Delta t' = t_2' - t_1' = \frac{\Delta t}{\sqrt{1-\frac{v^2}{c^2}}} \quad \text{……………………………(3)}$$

i.e., $\Delta t' > \Delta t$.

Now, if we consider the clock placed in S', then the clock is at rest relative to S' and is moving with velocity v relative to S. If it gives signal at t_1' and t_1 is the corresponding time in system S, then by inverse relations of Lorentz transformations, we have

$$t_1 = \frac{t_1' + \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (4)$$

If the clock again gives signal at t_2' and t_2 is the corresponding time in frame S, then

$$t_2 = \frac{t_2' + \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (5)$$

Therefore, from equations 4 and 5, the time interval will be,

$$t_2 - t_1 = \Delta t = \frac{t_2' - t_1'}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\therefore \Delta t = t_2 - t_1 = \frac{\Delta t'}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (6)$$

i.e. $\Delta t > \Delta t'$.

Hence, it is clear from equations 3 and 6 that the time interval registered by a clock appears to the moving observer to be delated or lengthened by a factor $\left(\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}\right)$.

Thus, from above reasoning we may say **"A moving clock always appears to go slow".**

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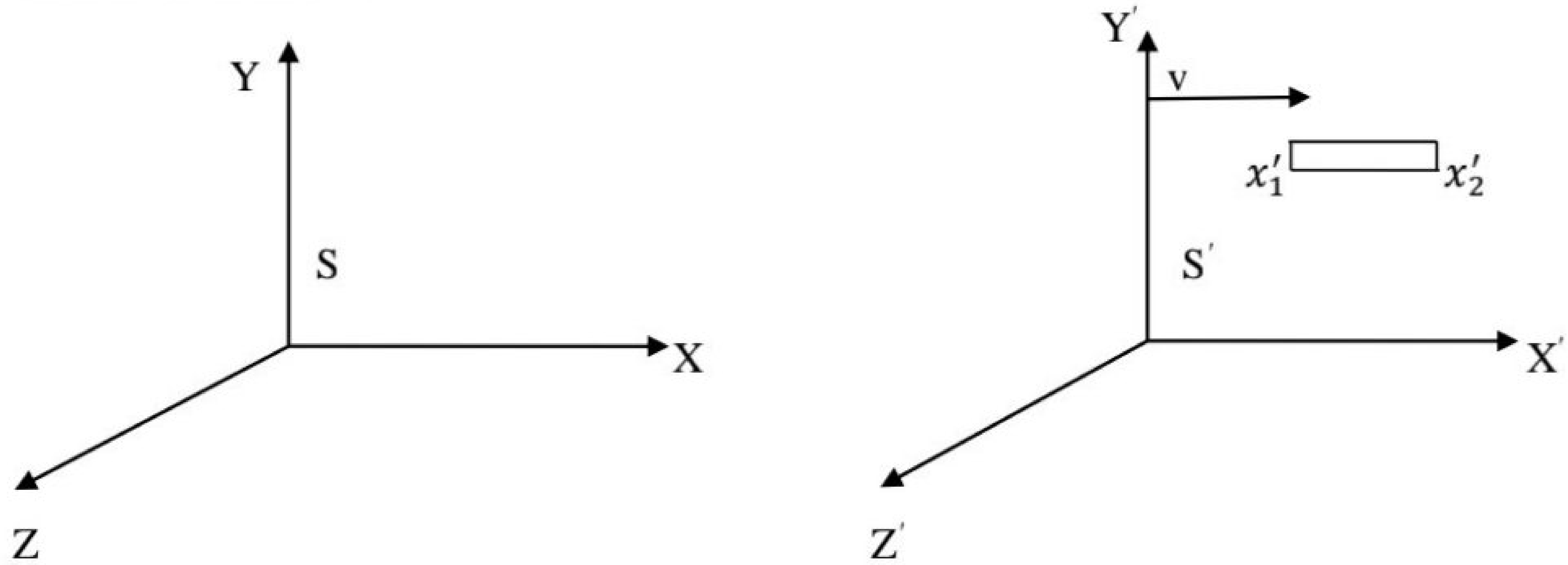
Length contraction:

Let us consider two frames of reference S and S'. S' is moving with velocity v relative to S along +ve direction of x-axis.



Let us consider a rod parallel to x-axis placed in system S', having the coordinates x_1 and x_2 in reference frame S. Then, the length of the rod in system S is,

$$L_o = x_2 - x_1 \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

Again, let an observer in the reference frame S' measure the end coordinates of the rod as x_1' and x_2'. Then the length of the rod in S' is,

$$L = x_2' - x_1' \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

According to inverse Lorentz transformation equations,

$$x_1 = \frac{x_1' + vt'}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (3.)$$

and $$x_2 = \frac{x_2' + vt'}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (4)$$

Therefore, from equation (1), using (3) and (4), we have,

$$L_o = x_2 - x_1 = \frac{x_2' - x_1'}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{L}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$L = L_o \sqrt{1 - \frac{v^2}{c^2}}$$

i.e., $L_o > L$

This follows that the length of rod seems to be contracted when measured by moving observer along the direction of length of rod in the ratio $\left[\mathbf{1}:\sqrt{\mathbf{1}-\frac{v^2}{c^2}}\right]$, v being the velocity of the observer.

Einstein's Mass-Energy relation:

From work-energy theorem, the kinetic energy of a moving body is equal to the work done by the external force that imparts the velocity to the body from rest, If $\bar{F}$ is the force acting on the body, then work done by the force on the body is rising its velocity from $v = 0$ to $v = v$ is given by,

$$W = \int_0^v \bar{F}.d\bar{S}$$

Kinetic energy of the body

$$E_k = T = W = \int_0^v \bar{F}.d\bar{S}$$

$$= \int_0^v \bar{F}.\frac{d\bar{S}}{dt}.dt$$

$$= \int_0^v \frac{d\bar{P}}{dt}.\bar{v}.dt$$

$$= \int_0^v \bar{v}.\frac{d}{dt}(m\bar{v}).dt$$

$$= \int_0^v \bar{v}.d(m\bar{v}) \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(1)$$

But, from the relation of variation of mass with velocity, the mass of body in motion

$m = \frac{m_o}{\sqrt{1-\frac{v^2}{c^2}}}$, m_o being rest mass of the body.

$$\therefore E_k = T = \int_0^v \bar{v}.d\left[\frac{m_o}{\sqrt{1-\frac{v^2}{c^2}}}\bar{v}\right]$$

$$= m_o \int_0^v \bar{v}.d\left[\bar{v}\left(1-\frac{v^2}{c^2}\right)^{-\frac{1}{2}}\right]$$

$$= m_o \int_0^v \bar{v}.\left[\left(1-\frac{v^2}{c^2}\right)^{-\frac{1}{2}} dv - v.\frac{1}{2}.\left(1-\frac{v^2}{c^2}\right)^{-\frac{3}{2}}.\left(-\frac{2v}{c^2}\right)dv\right]$$

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$$= m_o \int_0^v \bar{v}.\left(1-\frac{v^2}{c^2}\right)^{-\frac{3}{2}}\left[1-\frac{v^2}{c^2}+\frac{v^2}{c^2}\right]dv$$

$$= m_o \int_0^v \frac{\bar{v}dv}{\left(1-\frac{v^2}{c^2}\right)^{\frac{3}{2}}} \dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots.. (2)$$

Let, $y = 1 - \frac{v^2}{c^2}$
$\Rightarrow dy = -\frac{2v}{c^2}dv$

$\therefore vdv = -\frac{c^2 dy}{2}$
When $v = 0$, then $y = 1$
When $v = v$, then $y = 1 - \frac{v^2}{c^2}$
Therefore, from equation 2,

$$E_k = T = -\frac{m_o c^2}{2}\int_1^{1-\frac{v^2}{c^2}} \frac{dy}{y^{\frac{3}{2}}}$$

$$= m_o c^2 \left[y^{-\frac{1}{2}}\right]_1^{1-\frac{v^2}{c^2}}$$

$$= m_o c^2 \left[\left(1-\frac{v^2}{c^2}\right)^{-\frac{1}{2}} - 1\right]$$

$$= \left[\frac{m_o}{\sqrt{1-\frac{v^2}{c^2}}} - m_o\right]c^2$$

$$= (m - m_o)c^2 \dots\dots\dots\dots\dots\dots\dots\dots.(3)$$

This is the relativistic expression for kinetic energy.
Now, $m_o c^2$ may be regarded as rest energy of a body of rest mass m_o. Then total energy (E) of the body, i.e.,
$E = Rest\ energy + Relativistic\ energy$
$= m_o c^2 + (m - m_o)c^2$

$\boldsymbol{E = mc^2} \dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots (4)$

Which is known as Einstein's Mass-Energy relation.

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